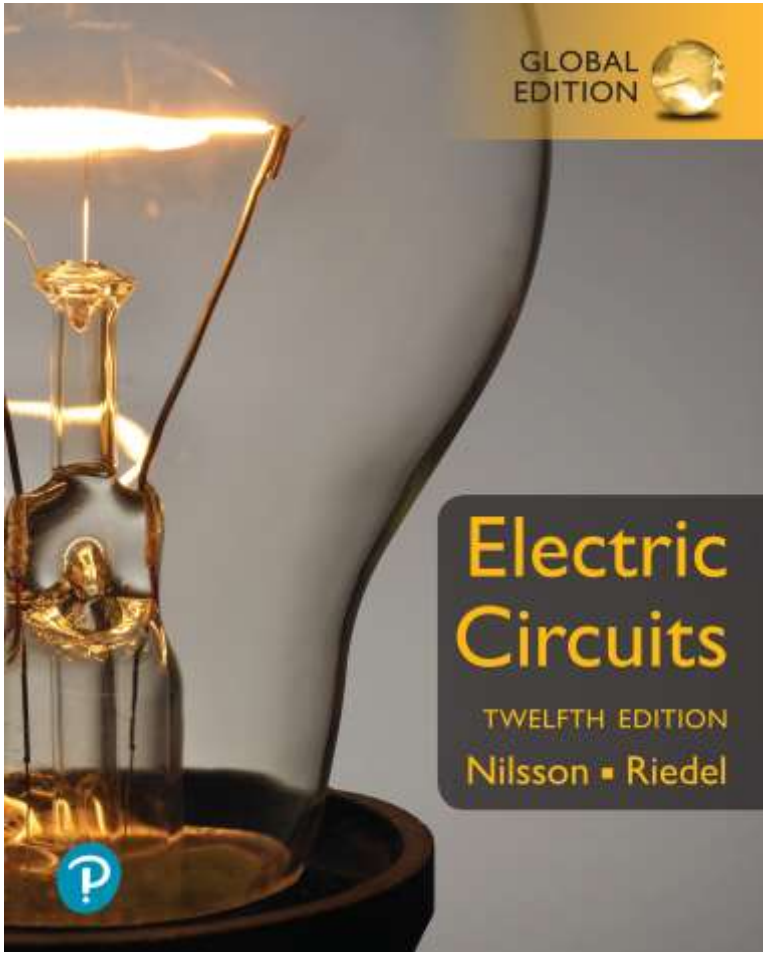
Electric Circuits

Twelfth Edition, Global Edition



Chapter 7 – Part 2

Response of First-Order *RL* and *RC* Circuits

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Chapter 7 – *RL* Natural Response

The analysis method:

Step 1 – Determine the initial inductor current, I_0, by analyzing the circuit for $t < 0$.

Step 2 – Calculate the time constant, τ = L/R, where R is the equivalent resistance connected to the inductor for $t \geq 0$.

Step 3 – Write the equation for the inductor current:

$$i(t) = I_o e^{-t/\tau}, \qquad t \geq 0.$$

Step 4 – Calculate other quantities of interest using the inductor current.

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Chapter 7 – *RC* Natural Response

The analysis method:

Step 1 – Determine the initial capacitor voltage, V_0, by analyzing the circuit for $t < 0$.

Step 2 – Calculate the time constant, $\tau = RC$, where R is the equivalent resistance connected to the capacitor for $t \geq 0$.

Step 3 – Write the equation for the capacitor voltage:

$$v(t) = V_o e^{-t/\tau}, \qquad t \geq 0.$$

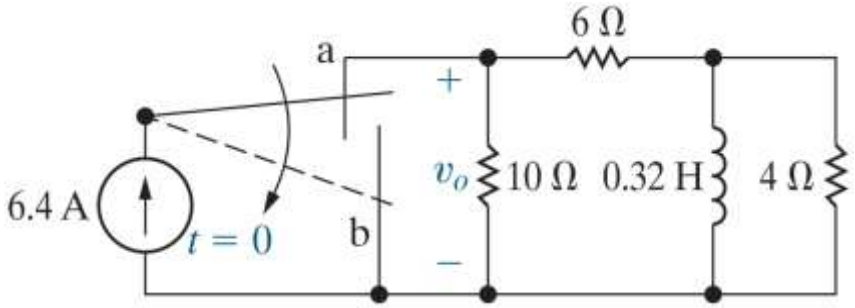
Step 4 – Calculate other quantities of interest using the capacitor voltage.

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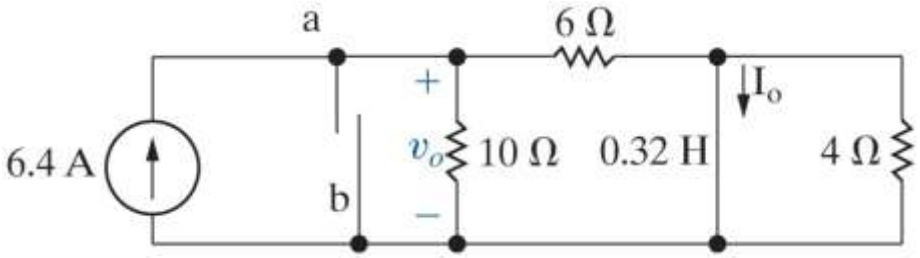
Chapter 7 – *RL/RC* Natural Response Examples (1 of 14)

Find $V_o(t)$ for $t \geq 0^+$



Step 1 – Determine the initial inductor current, I_0, by analyzing the circuit for $t < 0$.

For $t < 0$



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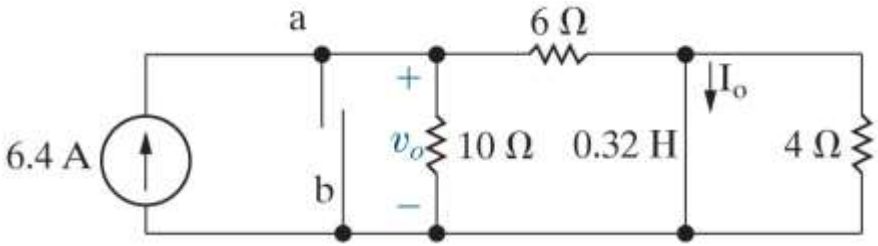
Chapter 7 – *RL/RC* Natural Response Examples (2 of 14)

The short circuit that replaced the inductor effectively eliminates which resistor from this circuit?

A. 10 Ω

B. 6 Ω

C. 4 Ω



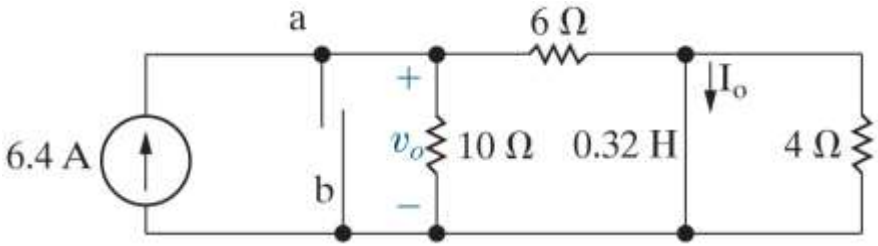
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Chapter 7 – *RL/RC* Natural Response Examples (2 of 14)

The short circuit that replaced the inductor effectively eliminates which resistor from this circuit?

- ✗ A. 10 Ω
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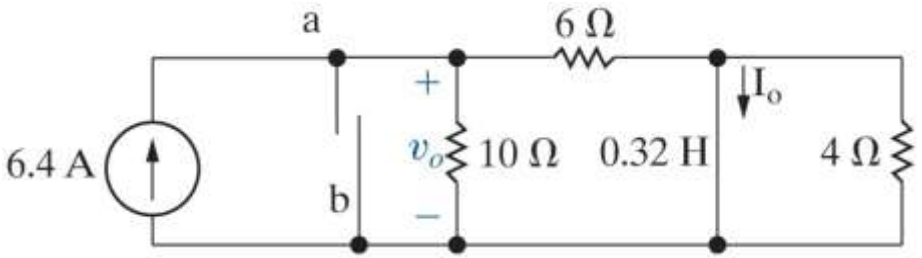


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Chapter 7 – *RL/RC* Natural Response Examples (3 of 14)

What circuit analysis tool provides the most direct method to calculate I_o?



A. Node voltage method

B. Current division

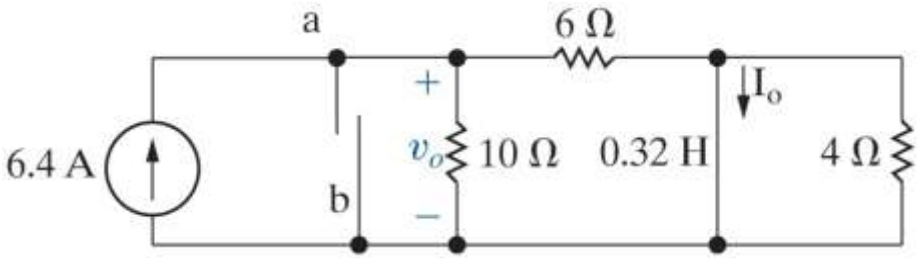
C. Source transformation

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Chapter 7 – *RL/RC* Natural Response Examples (3 of 14)

What circuit analysis tool provides the most direct method to calculate I_o?



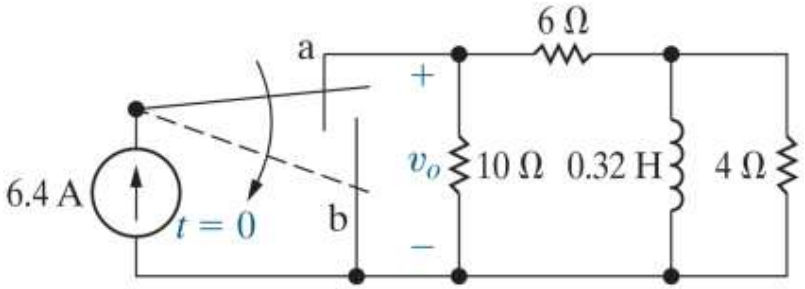
- ✗ A. Node voltage method
- ✓ B. Current division
- ✗ C. Source transformation

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Chapter 7 – *RL/RC* Natural Response Examples (4 of 14)

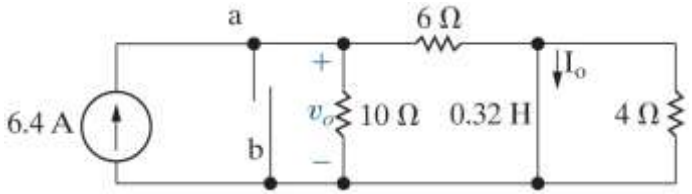
Find $v_o(t)$ for $t \geq 0^+$ (continued)



Step 1 – Determine the initial inductor current, I_0,

by analyzing the circuit for $t < 0$.

For $t < 0$



Current division:

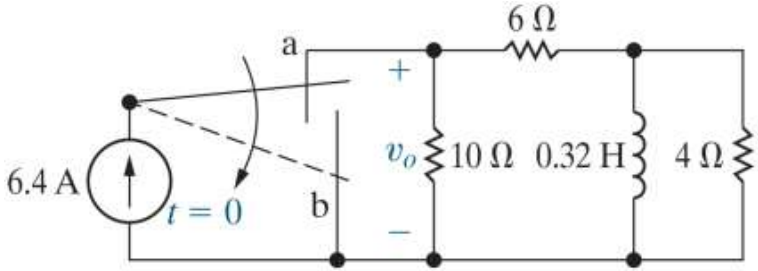
$$I_o = \frac{10}{10+6}(6.4) = 4 \text{ A}$$

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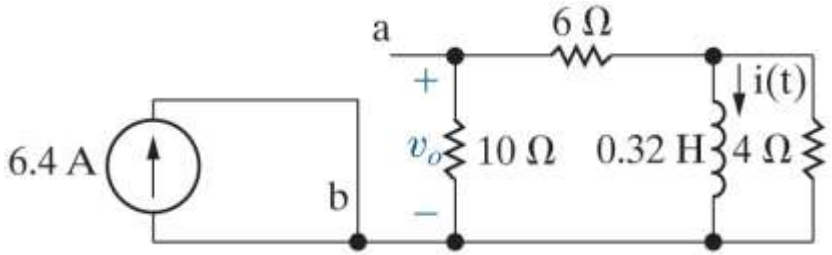
Chapter 7 – *RL/RC* Natural Response Examples (5 of 14)

Find $v_o(t)$ for $t \geq 0^+$ (continued)

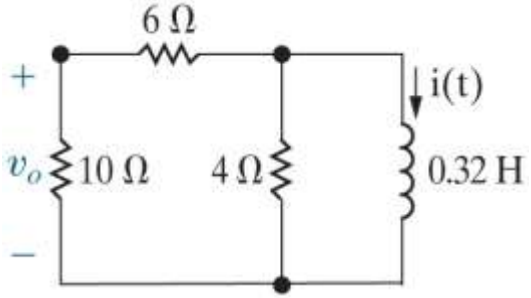


Step 2 – Calculate the time constant, $\tau = L/R$, where R is the equivalent resistance connected to the inductor for $t \geq 0$.

For $t \geq 0$



Simplified:



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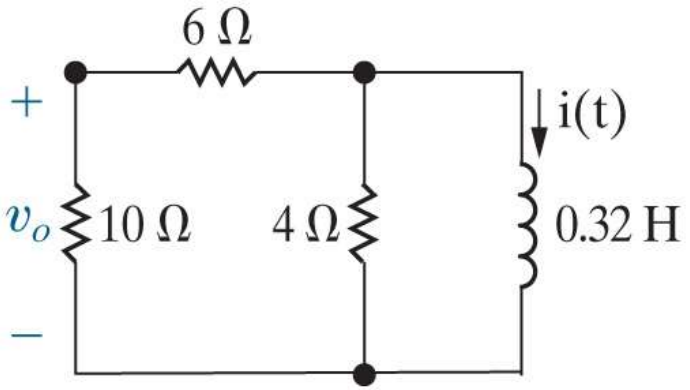
Chapter 7 – *RL/RC* Natural Response Examples (6 of 14)

The expression for the equivalent resistance seen by the inductor is

A. $4 \square (10 + 6)$

B. $10 \square (6 + 4)$

C. $10 + 6 + 4$



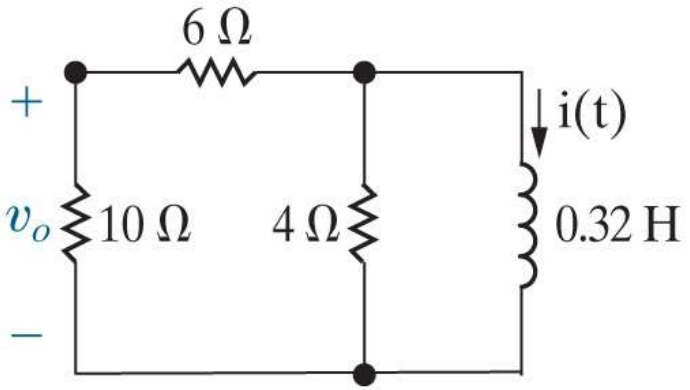
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Chapter 7 – *RL/RC* Natural Response Examples (6 of 14)

The expression for the equivalent resistance seen by the inductor is

- ✓ A. $4 \square (10 + 6)$
- ✗ B. $10 \square (6 + 4)$
- ✗ C. $10 + 6 + 4$

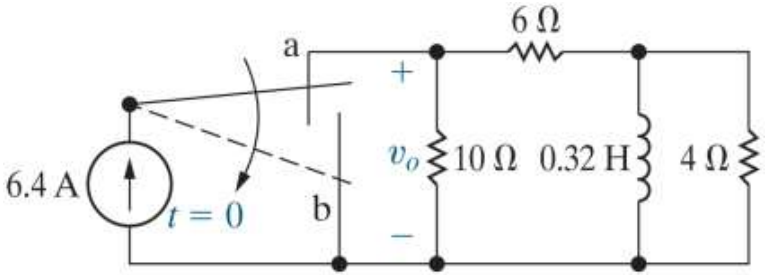


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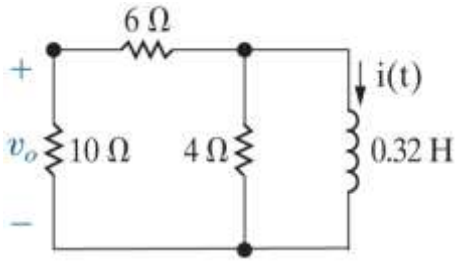
Chapter 7 – *RL/RC* Natural Response Examples (7 of 14)

Find v_o for $t \geq 0^+$ (continued)



Step 2 – Calculate the time constant, $\tau = L / R$, where R is the equivalent resistance connected to the inductor for $t \geq 0$.

For $t \geq 0$



$$R_{eq} = 4 \,||\, (6 + 10) = 3.2\Omega$$

$$\tau = L / R_{eq} = 0.32 / 3.2 = 0.1\,\text{s}$$

Step 3 – Write the equation for the inductor current:

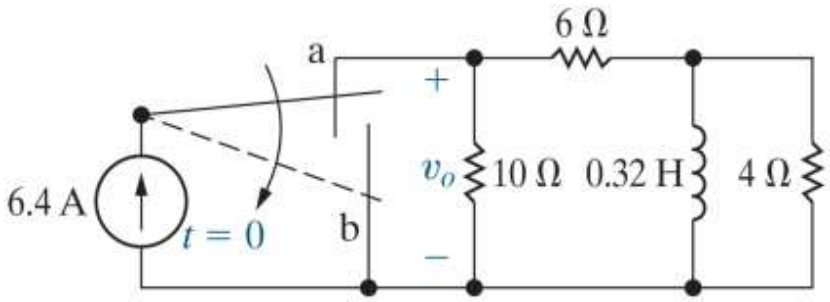
$$i(t) = I_o e^{-t/\tau} = 4e^{-t/0.1} = 4e^{-10t}\text{A}, \quad t \geq 0.$$

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Chapter 7 – *RL/RC* Natural Response Examples (8 of 14)

Find $v_o(t)$ for $t \geq 0^+$ (continued)



Step 4 – Calculate other quantities of interest using the inductor current. For this circuit, find the voltage across the inductor, positive at the top, and use voltage division to find v_o from the inductor voltage.

$$v_L(t) = (0.32)\frac{d}{dt}(4e^{-10t}) = (0.32)(-10)(4e^{-10t})$$

$$= -12.8e^{-10t} \text{ V}, \quad t \geq 0^+$$

$$v_o(t) = \frac{10}{10+6}v_L(t) = -8e^{-10t} \text{ V}, \quad t \geq 0^+$$

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Chapter 7 – *RL/RC* Natural Response Examples (9 of 14)

In circuits with inductors, we solve for the **inductor current** and the solution is valid for $t \geq 0$. When we use the inductor current to solve for any other voltage or current in the circuit, the resulting expression is valid for $t \geq 0^+$

A. True

B. False

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Chapter 7 – *RL/RC* Natural Response Examples (9 of 14)

In circuits with inductors, we solve for the **inductor current** and the solution is valid for $t \geq 0$. When we use the inductor current to solve for any other voltage or current in the circuit, the resulting expression is valid for $t \geq 0^+$

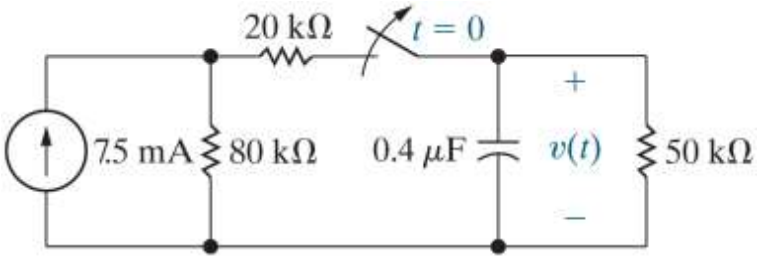
- ✔ A. True
- ✘ B. False

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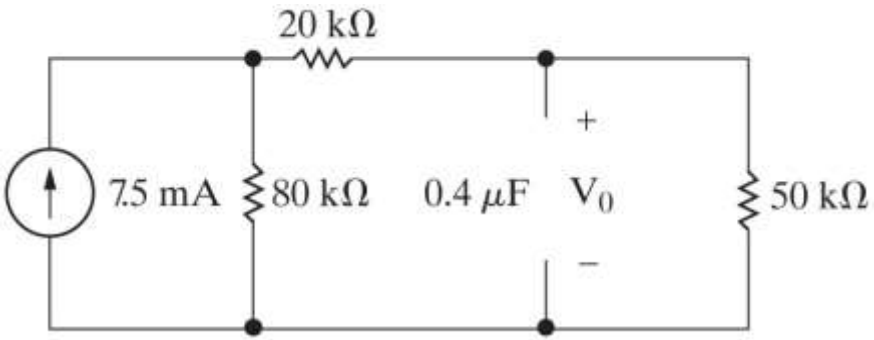
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Chapter 7 – *RL/RC* Natural Response Examples (10 of 14)

Find the time at which 75% of the initial stored energy has been dissipated.



Step 1 – Determine the initial capacitor voltage, V_0, by analyzing the circuit for $t < 0$.

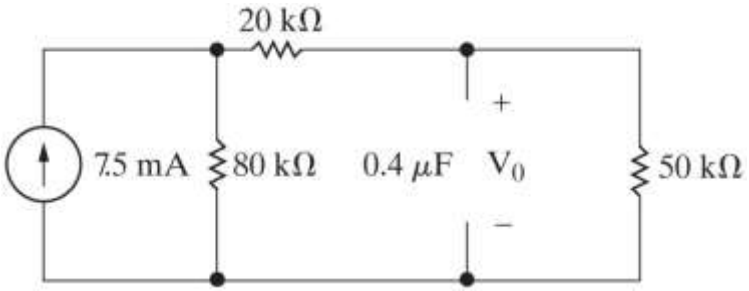


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Chapter 7 – *RL/RC* Natural Response Examples (11 of 14)

In the circuit used to find the initial voltage drop across the capacitor, the voltage drop across the open circuit is the same as the voltage drop across



A. The 80 KΩ resistor

B. The 50 KΩ resistor

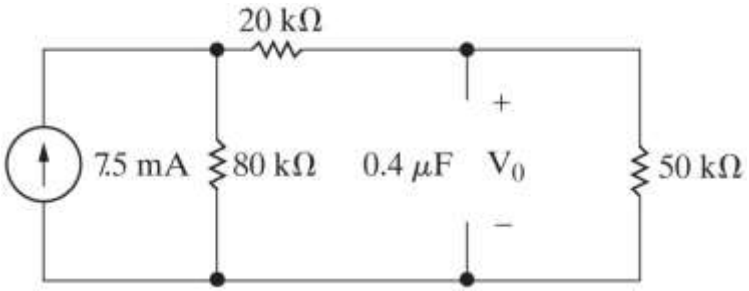
C. The 20 KΩ resistor

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Chapter 7 – *RL/RC* Natural Response Examples (11 of 14)

In the circuit used to find the initial voltage drop across the capacitor, the voltage drop across the open circuit is the same as the voltage drop across



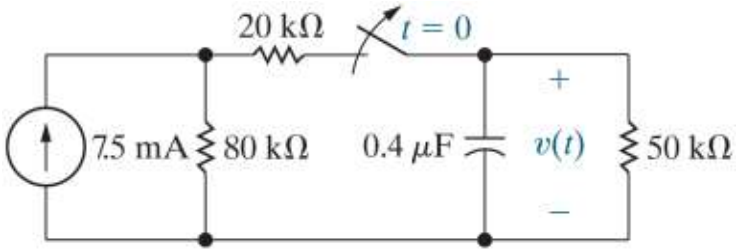
- ✗ A. The 80 KΩ resistor
- ✓ B. The 50 KΩ resistor
- ✗ C. The 20 KΩ resistor

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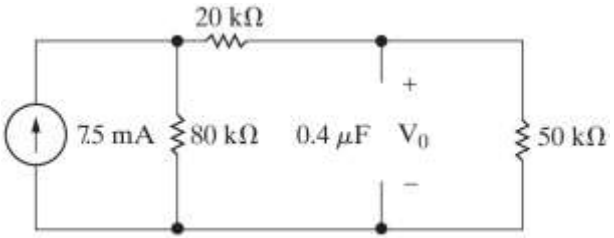
Chapter 7 – *RL/RC* Natural Response Examples (12 of 14)

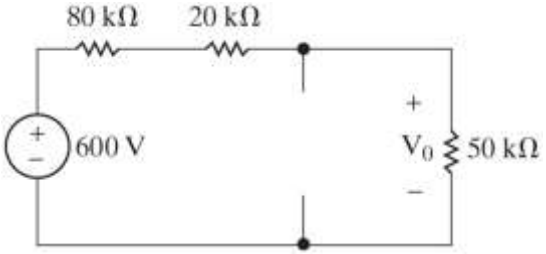
Find the time at which 75% of the initial stored energy has been dissipated (continued).



Step 1 – Determine the initial capacitor voltage, V_0, by analyzing the circuit for $t < 0$.

$t < 0$:





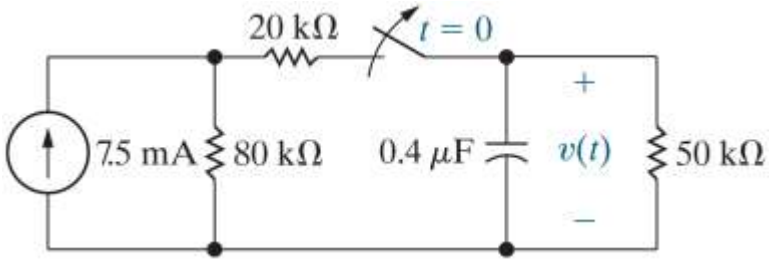
$$V_0 = \frac{50\text{ k}}{80\text{ k} + 20\text{ k} + 50\text{ k}}(600)$$
$$= 200\text{ V}$$

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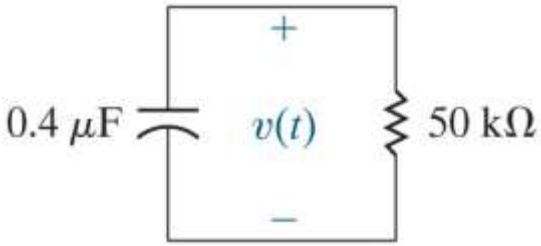
Chapter 7 – *RL/RC* Natural Response Examples (13 of 14)

Find the time at which 75% of the initial stored energy has been dissipated (continued).



Step 2 – Calculate the time constant, $\tau = RC$, where R is the equivalent resistance connected to the capacitor for $t \geq 0$.

$t \geq 0$:



$$\tau = (50{,}000)(0.4 \times 10^{-6}) = 0.02 \text{ s}$$

Step 3 – Write the equation for the capacitor voltage:

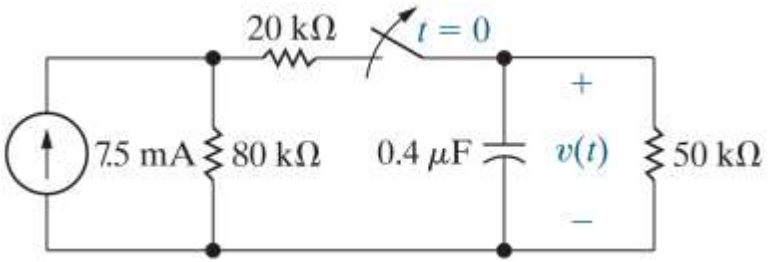
$$v(\text{t}) = V_o e^{-t/\tau} = 200\text{e}^{-t/0.02} = 200\text{e}^{-50\text{t}} \text{ V}, \quad \text{t} \geq 0.$$

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Chapter 7 – *RL/RC* Natural Response Examples (14 of 14)

Find the time at which 75% of the initial stored energy has been dissipated (continued).



$$v(t) = 200e^{-50t} \text{ V}, \quad t \geq 0$$

Step 4 – Calculate other quantities of interest using the capacitor
To start, calculate the initial stored energy in the capacitor.

$$w(0) = \tfrac{1}{2}(0.4 \times 10^{-6})(200)^2 = 8 \text{ mJ}$$

75% of the initial energy is 6 mJ, so the energy that remains is 2 mJ. Thus,

$$w(t) = \tfrac{1}{2}(0.4 \times 10^{-6})(200e^{-50t})^2 = 8e^{-100t} \text{ mJ} = 2 \text{ mJ}$$

$$\Rightarrow e^{-100t} = 0.25 \qquad \Rightarrow t = \frac{\ln(0.25)}{-100} = 13.86 \text{ ms}$$

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